Technology Stack

Date	28 June 2026
Team ID	
Project Name	Opti crop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*HTML, CSS, JavaScript*	*Used to create a responsive and user-friendly interface for entering soil parameters and displaying crop prediction results.*

2	**Backend / Server-Side**	*Python, Flask*	*Flask is used to build the web application and connect the machine learning model with the frontend for real-time crop prediction.*
3	**Database / Data Storage**	*MySQL*	*Stores user information, soil data, crop details, prediction history, and reports in a structured relational database.*
4	**Cloud / Hosting / Deployment**	*Localhost*	*Used to run and test the application during development. Can later be deployed to a cloud platform for public access.*
5	**Version Control & CI/CD**	***Github***	*Used to manage source code, collaborate with team members , and maintain different versions of the project.*

6	**Third-Party APIs / Other Tools**	*Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn*	*Used for data preprocessing, machine learning model training, numerical computation, and visualization of agricultural data.*